Vivekanand Education Society's Institute of Technology

Department of Computer Engineering

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YEH PROJECT UTHAYA GAYA HAI

KrishiMitra – AI-Powered Agricultural Assistant

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Abstract:

KrishiMitra is an AI-powered agricultural assistant developed to address critical farming challenges such as low crop yield, pest and disease outbreaks, lack of timely guidance, and limited market access. It leverages machine learning to provide personalized crop recommendations by analyzing soil type, climatic conditions, regional factors, and seasonal patterns, enabling farmers to make data-driven cultivation choices. The platform features a deep learning–based image analysis module for real-time plant disease detection, offering accurate identification and tailored treatment suggestions to prevent crop losses. Additionally, an intelligent weekly crop planner delivers stage-specific guidance from seed selection and sowing to irrigation schedules, pest control, harvesting, and storage dynamically adapting to crop type, growth stage, and changing environmental conditions.

A multilingual Retrieval-Augmented Generation (RAG) chatbot, trained on agricultural knowledge and localized data, answers farmer queries with contextual accuracy, bridging the information gap for rural communities. Furthermore, the system integrates real-time market insights, demand trend analysis, and direct selling options to connect farmers with buyers, helping them secure better prices and reduce reliance on intermediaries. By combining AI-driven insights, automated planning, and accessible market connectivity, KrishiMitra empowers farmers to make informed decisions, improve productivity, enhance profitability, and adopt sustainable agricultural practices for long-term resilience.

Introduction:

Agriculture remains the primary livelihood for rural communities, yet farmers often struggle with low yields, crop diseases, unpredictable weather, and limited market access. Traditional methods rarely provide the timely, data-driven insights needed to address these challenges. AI-powered solutions offer an opportunity to transform farming practices through precision, automation, and accessibility.

KrishiMitra is designed as an intelligent agricultural assistant that delivers personalized crop recommendations based on farmer-provided soil parameters such as nitrogen, pH level, and moisture content, along with climate and seasonal data. It integrates real-time plant disease detection via deep learning image analysis, a dynamic weekly crop planner, and a multilingual Retrieval-Augmented Generation (RAG) chatbot for accurate agricultural guidance.

The platform also provides market insights and direct selling opportunities, enabling farmers to make informed decisions, secure better prices, and adopt sustainable farming practices. By combining precision agriculture with AI-driven advisory tools, KrishiMitra aims to improve productivity, reduce losses, and enhance farmer profitability.

Furthermore, the system is designed to be accessible even in rural and resource-limited settings, ensuring that farmers with minimal technical skills can easily interact with its features through a simple, multilingual interface. This inclusivity ensures that modern agricultural innovations reach those who need them most, bridging the gap between advanced technology and grassroots farming.

Problem Statement

Agricultural communities, particularly in rural India, continue to face persistent challenges including low crop productivity, delayed or inadequate access to expert advice, undetected plant diseases, and insufficient awareness of market trends. The reliance on traditional methods and the lack of personalized, timely interventions contribute to reduced yield and financial instability. Existing solutions often fail to cater to farmers' linguistic diversity, technological literacy, and real-time decision-making needs. Moreover, most platforms operate in isolation, lacking integration of advanced AI capabilities for comprehensive support. Therefore, there is a pressing need for an intelligent, multilingual, and accessible system that leverages machine learning, deep learning, and retrieval-augmented natural language models to assist farmers in making informed, sustainable, and profitable agricultural decisions.

Proposed Solution :

KrishiMitra is conceptualized as an advanced AI-powered agricultural assistant that leverages modern machine learning, deep learning, and natural language processing techniques to address the multi-dimensional challenges faced by farmers. The proposed system integrates the following key components:

1. **Personalized Crop Recommendation System** – Utilizes supervised machine learning algorithms to analyze user-provided soil parameters (such as nitrogen content, pH level, and moisture percentage), along with real-time weather conditions and seasonal patterns. The model identifies and recommends the most suitable crops for cultivation, thereby maximizing yield potential and resource efficiency.

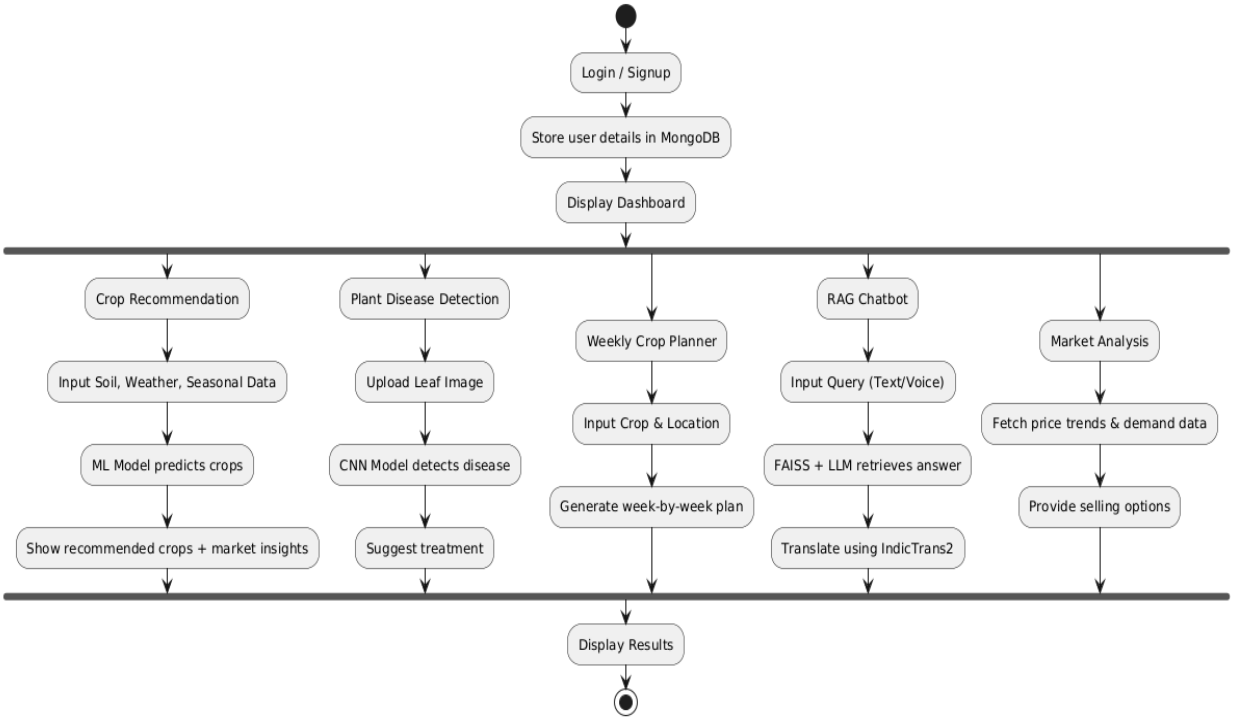
2. **Real-Time Plant Disease Detection and Advisory** – Employs deep convolutional neural networks (CNNs) for image classification to detect plant diseases from leaf images uploaded by farmers. Upon identification, the system suggests evidence-based treatment methods and preventive measures, reducing the risk of yield loss.

3. **Dynamic Weekly Crop Planning Module** – Generates an adaptive, stage-specific crop management plan that includes seed selection, irrigation scheduling, pesticide application timelines, and harvesting guidance. The plan is continuously updated based on crop growth stage and prevailing environmental conditions.

4. **Multilingual Retrieval-Augmented Generation (RAG) Chatbot** – Incorporates a large language model (LLM) enhanced with FAISS-based vector storage and semantic embeddings to retrieve contextually relevant agricultural information. The chatbot is capable of interacting with farmers in multiple regional languages, offering accurate, situation-specific responses to their queries.

5. **Market Analysis and Direct Selling Platform** – Provides real-time commodity price trends, demand forecasts, and an integrated marketplace for farmers to directly connect with buyers. This reduces dependency on intermediaries, ensuring better profit margins.

Block Diagram :



Methodology:

1. **User Access & Data Storage** – Farmers register or log in, providing farm, soil, and location details. Data such as nitrogen content, pH, and moisture levels are stored securely for generating personalized insights.

2. **Dashboard & Service Selection** – A simple, centralized dashboard lists all modules, including crop recommendation, plant disease detection, chatbot support, and market analysis.

3. **Crop Recommendation –** Machine learning models analyze soil parameters, local climate, and seasonal data to suggest profitable crops, linked with market demand and price trends.

4. **Plant Disease Prediction** – Farmers upload plant images, which are analyzed by deep learning models to detect diseases and suggest targeted treatments, with vendor connections if needed.

5. **Weekly Crop Planner –** Based on crop and location, the system generates a stage-wise farming calendar covering irrigation, fertilization, pest control, and harvesting.

6. **Multilingual RAG-Based Chatbot** – A Retrieval-Augmented Generation chatbot answers queries in regional languages with voice and translation support for accessibility.

7. **Market Analysis & Selling** – Real-time market insights help farmers set competitive prices and sell directly via the platform, reducing dependency on middlemen.

8. **Backend Intelligence –** AI/ML algorithms, backend services, and databases ensure smooth operation and allow future scalability for features like weather alerts or automation.

Hardware , Software and Tools Requirements:

- **Processor:** Intel i5
- **RAM:** 8 GB
- **Storage:** 256 GB SSD
- **GPU:** NVIDIA CUDA-enabled GPU
- **Languages:** Python, JavaScript,
- **Frameworks/Libraries:** React.js, Node.js, Flask, PyTorch, OpenCV, FAISS,Langchain
- **Database:** MongoDB

Proposed Evaluation Measures:

1. **Crop Recommendation Accuracy** – The system's crop suggestions will be compared against recommendations from agricultural experts and research-based guidelines to measure correctness and practical relevance.

2. **Disease Detection Performance** – Precision, recall, and F1-score will be calculated using a validated dataset of plant images to assess how effectively the deep learning model identifies and classifies crop diseases.

3. **Chatbot Quality** – The RAG-based chatbot will be evaluated for response relevance, contextual accuracy, and language clarity, supported by user satisfaction surveys and expert reviews.

4. **Weekly Plan Reliability** – The generated crop calendar will be reviewed by agricultural professionals to ensure alignment with region-specific best farming practices and seasonal requirements.

5. **Market Analysis Accuracy** – The accuracy of price and demand trends will be assessed by comparing platform-generated insights with real-time government and market data.

6. **Usability & Response Time** – System usability will be measured using the System Usability Scale (SUS), while query response and processing speeds will be tracked to ensure a smooth user experience.

7. **Adoption Rate** – The percentage of active farmers using the platform over time will be monitored to gauge acceptance, scalability, and long-term impact.

Conclusion :

KrishiMitra is an AI-powered agricultural assistant designed to tackle some of the most pressing issues faced by farmers, including optimal crop selection, timely disease detection, and improved market access. By integrating advanced machine learning algorithms for crop recommendations, deep learning models for plant disease detection, and a Retrieval-Augmented Generation (RAG)-based conversational chatbot, the platform delivers highly accurate, personalized, and timely guidance. The recommendations are generated using real-time inputs such as soil nutrient data, pH level, moisture content, climatic conditions, and seasonal patterns, ensuring they are context-specific and relevant to each farmer's unique needs.

In addition, KrishiMitra's multilingual support and intuitive interface make it highly accessible to farmers in rural communities, overcoming language and technology barriers. The platform not only assists in daily farming decisions through stage-wise crop planning but also offers real-time market insights, price trend analysis, and direct selling options to improve profitability. By enabling data-driven, sustainable, and efficient agricultural practices, KrishiMitra aims to increase crop yield, minimize losses, and enhance the socio-economic well-being of farmers, contributing to the modernization and resilience of the agriculture sector.

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